

Yuma Oda

Research Engineer | ML Systems, Compilers, Inference

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Second-year undergraduate at Keio University and Research Engineer at Mirai Labs, focused on ML systems, compiler infrastructure, and efficient LLM inference. Current research spans speculative decoding, model-runtime co-design, and neural network quantization. Long-standing interests include compilers and Common Lisp.

EDUCATION

Keio University — B.A. Candidate, Environment and Information Studies

Apr 2025 - Apr 2029 (Expected)

Focus: Computer Science, Transactions, and Databases

EXPERIENCE

Mirai Labs — Research Engineer

Mar 2026 - Present

Full-time · Remote · San Francisco, CA, USA

- Research model- and system-level methods for speculative decoding, including co-design of draft models and runtime verification, as described in this [paper](#).

Aoha, Inc. — Software Engineer

Dec 2023 - Aug 2026

Part-time · Hybrid · Minato-ku, Tokyo, Japan

- Built deep-learning compiler infrastructure and neural network quantization methods for computer vision models.
- Researched efficient training methods for mixture-of-experts models.
- Built reinforcement learning infrastructure across 56 NVIDIA H200 GPUs for automated RTL design, including agent architecture, training harnesses, and data/evaluation pipelines.

PUBLICATION

Yuma Oda, Ryan Mathieu, Roman Knyazhitskiy, and Artur Chakhvadze. “Trees from Marginals: Autoregressive drafting with factorized priors.” *arXiv preprint arXiv:2607.06763*, 2026.

PERSONAL PROJECT & OPEN SOURCE CONTRIBUTION

Caten — Creator and Maintainer

- Created an experimental deep-learning compiler in Common Lisp using lightweight IRs, polyhedral compilation, and an optimizing pattern matcher.
- Designed a RISC-style graph IR that lowers computation to 26 composable primitive operations.
- Built lazy evaluation, Clang JIT compilation, model inference demos, and experimental training and automatic differentiation support.

tinygrad — Fast-math approximations, [PR 5187](#)

- Completed an approximation bounty by implementing SIN, LOG2, and EXP2 approximations; merged into tinygrad.

HONORS & AWARDS

METI Minister’s Award, U-22 Programming Contest · \$3,600 award
Sokan no Susume Scholarship, Keio University · \$15,000 scholarship

Nov 2023

Apr 2025 - Apr 2029 (Expected)

LANGUAGES

Japanese — Native

English — Upper-intermediate (CEFR B2)